

# PENGYU JING

jingpengyu22@gmail.com

## EDUCATION

### National University of Singapore

PhD in Electrical and Computer Engineering

Singapore

August 2026 – Expected 2030

### National University of Singapore

M.Sc. in Robotics

Singapore

August 2025 – July 2026

- Cumulative GPA: 4.45/5.0

### University of Electronic Science and Technology of China (UESTC)

B.Eng. in Robotics Engineering

Chengdu, China

September 2021 – June 2025

- Cumulative GPA: 3.84/4.0

### University of Cambridge

Frontier Research Program in Deep Learning and Deep Neural Networks

Cambridge, UK

January 2024 – February 2024

## SKILLS

**Programming:** C/C++; Python; MATLAB

**Machine Learning:** PyTorch, reinforcement learning, vision-language-action models

**Robotics & Simulation:** ROS/ROS2, Isaac Sim, MuJoCo, LIBERO

**Hardware:** STM32, CAN, UART, SPI, PCB Design, Altium Designer, SolidWorks, 3D Printing

## PUBLICATIONS

### RARM: Confidence-Gated Progress Reward Modeling for RL in Manipulation.

Pengzhi Yang\*, Xinyu Wang\*, **Pengyu Jing\***, Kehan Wen\*, Yiduo Qu, Zhenhao Huang, Minghao Fu, Xin Liu, Yaheng Shen, Fan Shi. *CoRL*, 2026.

### Enabling Robust Cloth Manipulation via Inference-Time Simulator-in-the-Loop Refinement.

Xin Liu, Yulin Li, Ziming Li, **Pengyu Jing**, Zhenhao Huang, Bingyang Zhou, Ziqiu Zeng, Siyuan Luo, Chenkun Qi, Fan Shi. *CoRL*, 2026.

### DiagGen: Agentic Generation of Deformable Assets with Sim-based Diagnostics for Robotic Simulation.

Guanxiong Chen\*, Yiduo Qu\*, Qianjun Xia\*, **Pengyu Jing**, Yixian Cheng, Bole Ma, Pengzhi Yang, Bingyang Zhou, Ziming Li, Shashwat Suri, Gongbo Sun, Chao Liu, Peter Yichen Chen, Ziqiu Zeng, Fan Shi. *arXiv*, 2026.

### Agentic Real2Sim: Physics-based World Modeling with Vision-Language Agents.

Guanxiong Chen\*, Qianjun Xia\*, Jiawei Peng\*, Heng Zhang\*, **Pengyu Jing\***, Bole Ma\*, Justin Qian, Yixian Cheng, Ziyi Jiao, Bingyang Zhou, Yiduo Qu, Luoxin Ye, Kaifeng Zhang, Kunyi Wang, Weijia Zeng, Yunuo Chen, Pengzhi Yang, Ziqiu Zeng, Siyuan Luo, Huamin Wang, Chao Liu, Alan Yuille, Fan Shi, Changxi Zheng, Yunzhu Li, Chenfanfu Jiang, Peter Yichen Chen. *arXiv*, 2026.

### Design and Testing of a Pleurotus pulmonarius Stick Cutting Machine.

Chunlin Cai, **Pengyu Jing**, Lihong Wang, Qi Niu, Haobo Jiao, Chen Ma, Chongsong Li. *Agriculture*, 14(10), 1795, 2024.

### An Intermittent Fertilization Control System for Fruit Tree Crown Detection.

Hao Yin, **Pengyu Jing**, Chen Ma, Liewang Cao, Chongsong Li, Lihong Wang. *Agriculture*, 14(11), 1867, 2024.

## PROJECT EXPERIENCE

### ABU Robocon Elephant and Bunny Robots — National Champion Project

Electrical Group Leader (UESTC Robocon Team)

September 2022 – August 2023

- Led the electrical subsystem of a two-robot competition platform, owning the control architecture, sensing topology, and real-time communication design end to end.
- Architected a distributed real-time control network coordinating the central controller, motor drivers, and sensing nodes under hard latency budgets.
- Built a high-rate state-estimation pipeline fusing inertial and encoder measurements for closed-loop attitude and localization feedback on a fast-moving base.
- Developed a field-oriented actuation stack with cascaded torque, velocity, and position loops for high-bandwidth, precise motion tracking.
- Drove full-system integration, hardware-in-the-loop validation, and teleoperation interface design through to the national championship.

## HONORS & AWARDS

National Robotics Competition (Robocon) Champion

National Robotics Competition (Robocon) Best Design Award

ABU Asia-Pacific Robotics Competition (Robocon) Third Place

ABU Asia-Pacific Robotics Competition (Robocon) Best Design Award